

Practical no-4

Aim: Write a PL/SQL block to create a cursor to copy contents of one table into another. Avoid redundancy.

1. Problem Statement:

Cursors: (All types: Implicit, Explicit, Cursor FOR Loop, Parameterized Cursor)
Write a PL/SQL block of code using parameterized Cursor, that will merge the data available in the newly created table N_RollCall with the data available in the table O_RollCall. If the data in the first table already exist in the second table then that data should be skipped.

2. Learning Objectives

1. Understand the concept of cursors.
2. Understand implicit and explicit cursors.
3. Understand cursor-based row-by-row processing.
4. Implement a cursor inside a MySQL stored procedure.
5. Use a procedure parameter to control cursor processing.
6. Check for duplicate records.
7. Insert only new records into an existing table.

3. Hardware and Software requirements: requirements:

- Any CPU with Pentium Processor or similar, 256 MB
- RAM or more, 1 GB Hard Disk or more.
- Windows 7 Operating System, Oracle 11g, SQL developer

4. Theory:

A **cursor** is a database object used to process the rows returned by a query **one row at a time**. A cursor is a pointer to this context area. PL/SQL controls the context area through a cursor. A cursor holds the rows (one or more) returned by a SQL statement. The set of rows the cursor holds is referred to as the active set. You can name a cursor so that it could be referred to in a program to fetch and process the rows returned by the SQL statement, one at a time.

4.1 Types of Cursors

The practical syllabus mentions four types:

1. Implicit Cursor: -An implicit cursor is automatically created by the database when SQL statements such as **INSERT**, **UPDATE**, **DELETE**, or **SELECT** are executed. Implicit cursors are automatically created by Oracle whenever an SQL statement is executed, when there is no explicit cursor for the statement. Programmers cannot control the implicit cursors and the information in it.

Whenever a DML statement (INSERT, UPDATE and DELETE) is issued, an implicit cursor is associated with this statement. For INSERT operations, the cursor holds the data that needs to be inserted. For UPDATE and DELETE operations, the cursor identifies the rows that would be affected.

Example: UPDATE Student
SET Marks = 90
WHERE Student_ID = 101;

- The database internally manages the cursor.

2. Explicit Cursor:-An explicit cursor is declared and controlled by the programmer. Explicit cursors are programmer-defined cursors for gaining more control over the context area. An explicit cursor should be defined in the declaration section of the PL/SQL Block. It is created on a SELECT Statement which returns more than one row.

The general concept is:

-
- ```
graph TD; A[• DECLARE] --> B[• OPEN]; B --> C[• FETCH]; C --> D[• PROCESS]; D --> E[• CLOSE];
```
- DECLARE
  - ↓
  - OPEN
  - ↓
  - FETCH
  - ↓
  - PROCESS
  - ↓
  - CLOSE

**3. Cursor FOR Loop:-**In Oracle PL/SQL, a cursor **FOR LOOP** automatically opens, fetches and closes the cursor.

**Conceptually:**

- FOR record IN cursor\_name LOOP
- ...
- END LOOP;
- **Important:** MySQL does **not** provide Oracle's exact cursor **FOR LOOP** syntax. In MySQL, an explicit cursor with **OPEN**, **FETCH**, **LOOP**, and **CLOSE** is used instead.

**4. Parameterized Cursor:-**A parameterized cursor uses a parameter to determine which records are processed. A parameterized cursor is a cursor whose query depends on

a value supplied at runtime. A parameterized cursor allows us to pass a value to control which records the cursor will process.

## 2. Why Do We Need a Parameterized Cursor?

Consider a table containing 1000 records. Suppose we want to process records according to different departments.

| Without a parameter:     | With a parameter:                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process all 1000 records | Computer → Process only Computer records<br><br>IT → Process only IT records<br><br>AI&DS → Process only AI&DS records<br><br>Therefore, parameterized cursors make cursor processing flexible and reusable. |

**Conclusion:**

Thus we have successfully implemented PL/SQL block to create cursor to copy contents of one table into another